



AMITY UNIVERSITY

AMITY INSTITUTE OF INFORMATION TECHNOLOGY

DAILY DIARY

WPR of Week: 01

Date: 24th May, 2026

Program: MCA

NAME: DHRUV DHAYAL

ENROLLMENT NO.- A010145025215

PROJECT TITLE: RADMAP – GPS Enabled Radiation Mapping Prototype using Raspberry Pi

Weekly's Summary

Monday	Joined the AINST research laboratory and received an overview of the RADMAP project, its objectives, workflow, and applications in radiation monitoring.
Tuesday	Studied the fundamentals of radiation detection, Gamma Spectrometer operation, GPS technology, Raspberry Pi-5, and Digital Twin concepts through research papers and technical documentation.
Wednesday	Understood the overall hardware architecture and identified the components required for developing the radiation mapping prototype.
Thursday	Configured the Raspberry Pi-5 development environment, installed the required software libraries, and verified hardware communication.
Friday	Explored GPS communication protocols (NMEA) and understood how radiation data would be synchronized with geographical coordinates.
Saturday	Prepared the project implementation roadmap and discussed weekly milestones with the project mentor.



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WPR of Week: 02

Date: 27th May, 2026

Program: MCA

NAME: DHRUV DHAYAL

ENROLLMENT NO.- A010145025215

PROJECT TITLE: RADMAP – GPS Enabled Radiation Mapping Prototype using Raspberry Pi

Weekly's Summary

Monday	Integrated the GlobalSat BU-353N GPS Receiver with Raspberry Pi-5 and established serial communication.
Tuesday	Successfully acquired real-time latitude, longitude, UTC time, satellite information, and positioning accuracy from the GPS receiver.
Wednesday	Developed Python scripts to continuously capture GPS telemetry and store the data in structured format.
Thursday	Integrated environmental sensor readings including temperature and humidity with GPS data acquisition.
Friday	Performed multiple outdoor tests to verify GPS stability, positioning accuracy, and real-time synchronization.
Saturday	Prepared the GPS data acquisition module for further integration with radiation measurements.



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WPR of Week: 03

Date: 5th June, 2026

Program: MCA

NAME: DHRUV DHAYAL

ENROLLMENT NO.- A010145025215

PROJECT TITLE: RADMAP – GPS Enabled Radiation Mapping Prototype using Raspberry Pi

Weekly's Summary

Monday	Started collecting radiation and environmental datasets for visualization and mapping purposes.
Tuesday	Developed and organized the RADMAP dataset for further processing and machine learning experiments.
Wednesday	Uploaded the processed dataset to Kaggle for centralized storage, accessibility, and future research.
Thursday	Designed the initial 2D radiation heatmap using GPS coordinates and radiation count values.
Friday	Improved heatmap visualization using multiple color gradients to represent different radiation intensity levels.
Saturday	Validated the generated heatmaps using sample datasets and optimized the visualization pipeline.



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WPR of Week: 04

Date: 11th June, 2026

Program: MCA

NAME: DHRUV DHAYAL

ENROLLMENT NO.- A010145025215

PROJECT TITLE: RADMAP – GPS Enabled Radiation Mapping Prototype using Raspberry Pi

Weekly's Summary

Monday	Integrated the CNSPEC Multi-Channel Analyzer (Gamma Spectrometer) with the Raspberry Pi-5 backend system.
Tuesday	Performed radiation detection experiments using the Cobalt-60 (Co-60) radioactive source.
Wednesday	Conducted radiation testing using the Cesium-137 (Cs-137) source and analyzed the obtained spectra.
Thursday	Validated detector calibration, count rates, and peak channel measurements under different conditions.
Friday	Integrated Gamma Spectrometer output with GPS and environmental data for synchronized mapping.
Saturday	Successfully completed dynamic radiation acquisition and verified the overall detector performance.



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WPR of Week: 05

Date: 16th June, 2026

Program: MCA

NAME: DHRUV DHAYAL

ENROLLMENT NO.- A010145025215

PROJECT TITLE: RADMAP – GPS Enabled Radiation Mapping Prototype using Raspberry Pi

Weekly's Summary

Monday	Developed the backend system for real-time acquisition of radiation, GPS, and environmental data.
Tuesday	Integrated all hardware components including Raspberry Pi-5, GPS receiver, sensors, and Gamma Spectrometer into a unified architecture.
Wednesday	Implemented synchronized data processing for continuous radiation monitoring.
Thursday	Developed visualization modules to display radiation counts, peak channels, and GPS-tagged measurements.
Friday	Optimized backend performance for real-time processing and stable communication among hardware modules.
Saturday	Conducted integration testing and verified the complete backend workflow.



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WPR of Week: 06

Date: 28th June, 2026

Program: MCA

NAME: DHRUV DHAYAL

ENROLLMENT NO.- A010145025215

PROJECT TITLE: RADMAP – GPS Enabled Radiation Mapping Prototype using Raspberry Pi

Weekly's Summary

Monday	Identified the limitations of conventional 2D radiation heatmaps in locating radioactive sources.
Tuesday	Conducted research on Digital Twin technology and SLAM-based visualization techniques.
Wednesday	Designed the architecture for a real-time 3D Digital Twin using GPS and radiation data.
Thursday	Developed the initial 3D Digital Twin visualization integrated with real-time GPS coordinates.
Friday	Integrated live camera feedback with the Digital Twin model for improved spatial visualization.
Saturday	Evaluated the Digital Twin framework and documented its advantages for future source SLAM Model localization.



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WPR of Week: 07

Date: 1st July, 2026

Program: MCA

NAME: DHRUV DHAYAL

ENROLLMENT NO.- A010145025215

PROJECT TITLE: RADMAP – GPS Enabled Radiation Mapping Prototype using Raspberry Pi

Weekly's Summary

Monday	Developed a radiation color grading system inspired by night-vision imaging to visualize varying radiation intensity levels.
Tuesday	Implemented real-time radiation overlay on the live camera feed using dynamic color mapping techniques.
Wednesday	Enhanced the integrated system by synchronizing radiation data, GPS information, and camera visualization.
Thursday	Designed the complete RADMAP hardware prototype and prepared the structural model for 3D printing.
Friday	Performed final testing and validation of the complete RADMAP prototype, including 2D mapping, Digital Twin, and backend integration.
Saturday	Presented the developed prototype to Dr. Abhishek Yadav

	and the Nuclear Science research team. The project received positive evaluation and approval for future enhancement and funding support.
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